Sichuan University - Pittsburgh Institute

ECE 1212 **Electronic Circuit Design Lab** **Autumn 2026**

Note: This syllabus is subject to change during the semester. Any changes to the syllabus will be announced in class or posted on the Blackboard course area.

Introduction/Learning Objectives: This is a laboratory on electronic circuits experiments with an emphasis on circuit applications and design. Topics include multi-stage amplifier design; operational amplifier applications; analog-digital conversion applications, and active filters.

During this course, you are asked to design and perform seven experiments. Each experiment is introduced during a lecture given during the first portion of the lab. Usually, two students, forming a "group", share responsibility for the execution of the experiment during the lab periods. If extra time is required, the Laboratory is always available except during hours when other classes are in session.

Prerequisites: ECE 0101 and ECE 0102

Applicable ABET Outcomes:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
3. an ability to communicate effectively with a range of audiences.
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. an ability to acquire and apply new knowledge as needed, using T learning strategies.

Instructor: Dr. Lawrence Lam

Section 1: Lectures (N210) -Tuesday 8:15-9:55am; Lab-Tuesday: 10:15-11:55am

Section 2: Lectures (N210) -Wednesday 1:50-3:30pm; Lab-Wednesday: 3:40-5:30pm

Office Hours: Tue: 1:30-5:30pm; Wed: 9-11:30am;

Location: SCUPI Rm513

Email: lawrence.lam@scai-ji.cn **(Please include "ECE 1212" and some topic keywords in the subject line.)**

Teaching Assistants:

Theory TAs: Torres Liu and Rain Ye

Lab TA: Section 1- Bonnie Wang, Suzzie Zou, Section 2- Orient Wang, Mathew Zeng

Office hours and contact info, will be provided to you by the TAs in the QQ group

Reference Textbook: *Microelectronic Circuits* (7th Edition), Sedra, Smith, Carusone, Gaudet, Oxford University Press.

Grading:

Lab Prelab Plans	10%
Lab Final Reports	25%
Quizzes in class (theory)	10%
Performance Evaluation	5%
Midterm	25%
Final Project	25%

Letter grades will be determined from accumulated point totals and assigned according to the scale below.

A: 90 - 100	A-: 85 - 90	B+: 80 - 85	B: 76 - 80	B-: 73 - 76
C+: 70 - 73	C: 66 - 70	C-: 63 - 66	D: 60 - 63	F: < 60

Note: Up to 5% points could be granted to the final grade based on the overall course performance (lecture attendance, homework submission punctuality, course engagement such as in-class question answering and discussion, etc.).

During this course, you are asked to design and perform six experiments. Each experiment is introduced during a lecture given during the first portion of the lab. Usually, two students, forming a "group", share responsibility for the execution of the experiment during the lab periods. If extra time is required, the Laboratory is always available except during hours when other classes are in session.

Student endeavors will include:

1. Preparation of work plans (or Prelab) for each experiment (or portion of an experiment). These Prelabs should be based on independent reading, analysis and suggestions given in the lecture. In general, I will lecture on an experiment (or part of an experiment) before the prelab is due. Each group only needs to submit one copy of their pre-lab by one member of the group, however, the name of the other member must be specified in the Pre-lab. Prelabs should be graded by the Lab TAs before a group begins execution of the laboratory.

Pre-lab work generally consists of the derivation of design equations, designing the circuits you will later construct, and other types of analysis. Unless otherwise instructed, all circuit designs should use single components of standard values. For example, if your design calls for a resistor of 1050 ohms, you should use a single 1 kilohm resistor during construction. Do not use multiple components in series or parallel unless specifically instructed to do so.

2. Conducting the actual experiments in the laboratory. Each group or an individual student must have a laboratory notebook, which can be a written record or a computer file, to record data. Experimental results should be placed in the notebook. The instructor or TAs may examine the notebook during the lab period. **Some elements of each experiment must be demonstrated to the Instructor or TAs in the lab.** These elements may include live demonstrations of working circuits, showing the required measurements, oscilloscope displays, or other results.

3. Preparation of a Final Report for each experiment (one per group). Each group should submit their lab report by the due time. This report should be prepared separately

from the notebook and should be prepared on a word processor in MSWord and then saved in pdf file format to submit. Laboratory reports will be prepared for each full experiment and are usually due one week after the experiment is scheduled for completion in the laboratory.

Prelab and Lab Reports

1. A format for lab prelabs and reports will be provided later.
2. The due dates of prelabs and final reports will be posted as assignments.
3. Any prelab report or final report submitted after the due time will be considered late and see the late policy below.

Quizzes and Midterm Exam

- There is a quiz each week during lecture time and will cover material related to recently completed lecture or laboratory experiment.
- Attendance at the final exam is mandatory. An unexcused absence from this exam will result in a grade of 0.
- Midterm exam is a written exam and will base on lectures and experiments covered.

Final Design Project

There will be no final exam, instead, there will be a final design project (hands on, theory and lab design with data verification of the theory and design). The final design project topic will be given around week 13, and the final design project is due on week 16 with demonstration. The design project report is due at the end of week 16 Sunday 11:59pm. No late submission.

Late Work and Make-up Policy: Late homework submissions will be accepted up to 24 hours after the due date with a 20% penalty unless an extension request is approved **before the homework due date**. Extensions are granted at the instructor's discretion. In general, **no** make-up for quizzes and exams will be allowed. Exceptions will only be made for special circumstances such as a medical emergency. If you cannot attend a quiz or an exam, you must contact the instructor before the quiz and exam. Failure to do so will result in a zero grade on that quiz or exam.

Grade Rebuttal: For any quiz or exam, you have one week to request correction if you feel your answer might be mis-graded. No correction will be made a week after the quiz or exam grade is posted.

Disability Services: If you have a disability for which you are or may be requesting an accommodation, you are encouraged to contact the instructor as early as possible in the semester. Reasonable accommodations will be arranged for this course.

Communication with Instructor for Absence: In any situation regarding class absence, a student who becomes ill or has other emergency issues is responsible for communicating with the instructor. Please contact the instructor or Teaching Assistant via email before the lecture or lab to be involved.

Academic Integrity: Students in this course will be expected to comply with the SCUPI and/or the University of Pittsburgh's Policy on Academic Integrity. Any student suspected of violating this obligation for any reason during the semester will be required to participate in the procedural process, initiated at the instructor level, as outlined in the University Guidelines on Academic Integrity. This may include but is not limited to, the confiscation of the examination of any individual suspected of violating University Policy.

Plagiarism

- Students must do their own work. Plagiarism of another student's work is a serious offense and can result in dismissal from the University of Pittsburgh.
- Students in this course will be expected to comply with the University of Pittsburgh's Policy on Academic Integrity. Any student suspected of violating this obligation for any reason during the semester will be required to participate in the procedural process, initiated at the instructor level, as outlined in the University Guidelines on Academic Integrity. This may include, but is not limited to, the confiscation of the examination of any individual suspected of violating University Policy.

Attendance: Attendance is required for each lecture and lab as part of the performance evaluation. Taking the quiz in lecture and uploading the prelab in lab time as part of the performance.

Audio/Video Recording: To ensure free and open discussion, students may not record lectures, discussions, and/or any other class activities without advance written permission from the instructor, and any such recording properly approved in advance can solely be limited to the student's own private use.

Tentative Course Topics (subject to changes):

- Multi-stage amplifier design
- Operational amplifier applications
- Analog-digital conversions
- Active filters

Tentative Course Schedule (subject to change)

Week	ECE1212 (Electronic Circuit Design Lab)	Topics	Assignment
1	Lab1 Frequency Response of Opamp	Syllabus & Introduction to Opamps	Prelab1
2			
3	Lab2 Opamp Applications	Opamps Applications	Prelab2
4			
5			
6	Lab3 Active Filters & Oscillators	Active Filters	Prelab3
7			
8			
9	Lab4 Amplifier Design Using Bipolar Junction Transistor	Bipolar Junction Amplifier	Prelab4
10			
11	Lab5 MOSFET Amplifier & Current Mirrors	MOSFET Amplifier	Prelab5
12	Midterm & Final Project topic		
13		A/D Applications	Prelab6
14	Lab6 Analog-Digital Applications & Shift Registers applications		Lab6 due
15	Final Design Project		
16	Final project due		
17			
18			
19			
20			